

## 4.4 Segregation of the recessive by inbreeding

The frequency of the homozygotic recessive individuals in the population will increase by inbreeding, this is particularly true when the gene frequency is low.

**Example:** In a population the gene frequency of a recessive gene is 0.01 corresponding to a frequency of homozygote recessive individuals at 0.0001 or 1 in 10,000. If full sib mating forms the new generation, only, the inbreeding coefficient is 25% in all individuals in the next generation.

In this population the frequency of homozygote recessive individuals can be calculated as follows: It is proportional to the number of heterozygotes individuals in the parents' generation ( $2 \cdot 0,01 \cdot 0,99 = 0,0198$ ) and their chance that segregation of the recessive occurs ( $1/16$ ) by full sib mating (Figure 4.1), this segregation should equal one of the genes, the disease gene. Either the grandfather or the grandmother can carry the disease gene. Thus, the joint probability should be multiplied by two.

Therefore the joint probability of breeding homozygote recessive individuals is  $2 \cdot 0,0198 / 16 = 0,0025$ , which means that it has increased 25 times compared to the outbred base population. (In these calculations the case where other combinations of heterozygosity and homozygosity of the grandparents are not taken into account, so the result deviate slightly when compared to the general formulas shown below). If the formulas given below are used in our example, the result is  $q^2 + pqF = 0.0001 + 0.99 \cdot 0.01 \cdot 0.25 = 0.0026$ .

Correspondingly, the joint number of homozygote dominants will be increased by the same proportion as the recessive at the expense of the heterozygotic individuals. During inbreeding the proportions would change in comparison to the expected frequency under Hardy Weinberg equilibrium with the following expectations, exp:

|                |        |         |        |
|----------------|--------|---------|--------|
| -----          |        |         |        |
| Genotype       | AA     | Aa      | aa     |
| Frequency, exp | $p^2$  | $2pq$   | $q^2$  |
|                | $+pqF$ | $-2pqF$ | $+pqF$ |

By means of the expected frequencies the degree of inbreeding can be calculated in sub populations;

$$\text{which gives } F = (H_0 - H_n) / H_0 ,$$

where  $H_0$  and  $H_n$  are the genotype frequencies for heterozygotes in respectively the base population and generation n of the subdivided populations.

Applied on the Albumin example of dogs, section 2.4, the average inbreeding within dog breeds is  $F = (0.490 - 0.330) / 0.490 = 0.33$ .